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Connector connection structure

Abstract

The first connector includes a main body, a first connecting surface, a rotating ring, and a restricting pin for prohibiting the rotation of the rotating ring relative to the first connecting surface, the rotating ring, when taking a predetermined rotational phase with respect to the second connector, the second connector to the abutting position, can be drawn in the axial direction, further, the second connector is engaged with a portion of the second connector by rotating relative to the first connecting surface in a state of being retracted to the abutting position to maintain a connection state, the restricting pin, when the second connector is retracted to the abutting position, the rotation of the rotating ring from the restricting position to allow the rotation of the allowable position.

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Background/Summary**CROSS-REFERENCE TO RELATED APPLICATION**

(1) This application claims priority to Japanese Patent Application No. 2022-103762 filed on Jun. 28, 2022, incorporated herein by reference in its entirety.

BACKGROUND**1. Technical Field**

(2) The present specification discloses a connector connection structure that connects a first connector provided for one of a supply source and a reservoir of liquid hydrogen and a second connector provided for the other of the supply source and the reservoir.

2. Description of Related Art

(3) There may be a case where hydrogen is supplied from a supply source to a reservoir while in a liquid state. For example, in some hydrogen stations, hydrogen is received and stored in a liquid state from a tank truck that is a supply source. Typically, the supply source is provided with a first connector and the reservoir is provided with a second connector connectable to the first connector. Connecting the first connector and the second connector makes it possible to supply the liquid hydrogen from the supply source to the reservoir.

SUMMARY

(4) The first connector and the second connector need to be connected in a liquid-tight manner so as to appropriately supply the liquid hydrogen. In order to enable such a connection, in some cases, a technique has been proposed in which the first connector is screwed and fastened to the second connector in a state where the first connector is in close contact with the second connector. However, in this case, a connection operation becomes very complicated, and the connection operation is thus time consuming.

(5) Japanese Unexamined Patent Application Publication No. 2013-071778 (JP 2013-071778 A) discloses a technique for taking out a liquid stored in a tank. However, the technique disclosed in JP 2013-071778 A utilizes a siphon effect, and thus it is difficult to apply the technique to supply of the liquid hydrogen.

(6) Therefore, the present specification discloses a connection structure in which the first connector and the second connector used for supplying the liquid hydrogen can be connected more easily.

(7) A connector connection structure disclosed in the present specification is a connector connection structure that connects a first connector provided for one of a supply source and a reservoir of liquid hydrogen and a second connector provided for the other of the supply source and the reservoir.

(8) The first connector includes a main body, a first connecting surface that is fixed to the main

body and abuts a second connecting surface of the second connector, a rotating ring that surrounds the first connecting surface and is rotatable with respect to the first connecting surface, and a restricting pin that prohibits a rotation of the rotating ring with respect to the first connecting surface.

When the rotating ring takes a predetermined rotational phase with respect to the second connector, the rotating ring is able to pull in the second connector in an axial direction to an abutting position at which the second connecting surface abuts the first connecting surface, and further, rotates with respect to the first connecting surface in a state where the second connector is pulled in to the abutting position and partially engages with the second connector so as to maintain a connection state.

When the second connector is pulled in to the abutting position, the restricting pin moves from a restricting position at which the rotation of the rotating ring is prohibited to an allowable position at which the rotation of the rotating ring is allowed.

(9) According to the technique disclosed in the present specification, it is possible to easily perform a positioning operation of the rotational phase of the rotating ring with respect to the second connector and an engagement operation between the rotating ring and the second connector. As a result, according to the technique disclosed in the present specification, the first connector and the second connector used for supplying the liquid hydrogen can be connected more easily.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:

(2) FIG. 1 is a perspective view of a first connector and a second connector;

(3) FIG. 2 is a cross-sectional view of the first connector;

(4) FIG. 3 is a perspective view of a rotating ring;

(5) FIG. 4 is an enlarged view of the section A of FIG. 2 with the restricting pin in the restricting position;

(6) FIG. 5 is an A part enlarged view of the FIG. 2 in the state where a restricting pin is in an allowable position

(7) FIG. 6A is a view illustrating the interlocking member and the receiving recess when the first valve is closed;

(8) FIG. 6B is another view illustrating the positional relation of the interlocking member and the receiving recess with the first valve closed;

(9) FIG. 6C is another view illustrating the positional relation of the interlocking member and the receiving recess with the first valve closed;

(10) FIG. 7A is a view for explaining a positional relation between an interlocking member and a receiving recess in a state where a first valve is opened;

(11) FIG. 7B is another view illustrating the positional relation of the interlocking member and the receiving recess with the first valve open; and

(12) FIG. 7C, while opening the first valve is another view for explaining the positional relation of the interlocking member and the receiving recess.

DETAILED DESCRIPTION OF EMBODIMENTS

(13) Hereinafter, a connector connection structure will be described with reference to the drawings.

FIG. 1 is a perspective view of a first connector **10** and a second connector **70**. FIG. 2 is a cross-sectional view of the first connector **10**. The first connector **10** is connected to a supply source **100** of liquid hydrogen. The second connector **70** is connected to the liquid hydrogen reservoir **110**.

When the liquid hydrogen is supplied from the supply source **100** to the reservoir **110**, the first connector **10** and the second connector **70** are connected in a liquid-tight manner. If the reservoir **110** is a fuel cell vehicle or a hydrogen engine vehicle, the supply source **100** is a hydrogen tank of a hydrogen station or a tank lorry that transports liquid hydrogen. When the reservoir **110** is a hydrogen tank of a hydrogen station, the supply source **100** is a tank truck. Alternatively, the second connector **70** may be provided in the supply source **100** and the first connector **10** may be provided in the reservoir **110**.

(14) The first connector **10** includes a main body **12**. As shown in FIG. 2, the main body **12** has a radially overhanging flange portion **12a** at its distal end. A first flow path **16** is formed inside the main body **12**. The first flow path **16** includes a gas pipe **20** and a liquid pipe **18** disposed inside the gas pipe **20**. That is, the first flow path **16** has a double tube structure. The liquid hydrogen flows through the inside of the liquid pipe **18** to the second connector **70**. The vaporized hydrogen gas is sucked to the supply source **100** side through the gap between the liquid pipe **18** and the gas pipe **20**.

(15) As shown in FIG. 2, a first connecting plate **30** is provided at the end of the first connector **10** (i.e., the end on the second connector **70** side). The first connecting plate **30** is mechanically connected to the main body **12** via an intermediate tube **24**. Therefore, the first connector **10** does not rotate with respect to the main body **12**. The end surface of the first connecting plate **30** functions as a first connecting surface **32** to which a second connecting plate **72** to be described later abuts. A center hole **34** is formed in the center of the first connecting plate **30**. A first valve **26** is disposed between the center hole **34** and the first flow path **16**. The first valve **26** is a ball valve. The first valve **26** rotates about the rotation shaft **68** to thereby communicate or disconnect the center hole **34** and the first flow path **16**.

(16) The rotating ring **40** is a cylindrical member that surrounds the first connecting plate **30**. As shown in FIG. 1, the rotating ring **40** is mechanically coupled to the rotating body **46** via a connecting bar **52**. As shown in FIGS. 1 and 2, the rotating body **46** is a tubular member that surrounds the periphery of the main body **12**. A handle **50** gripped by an operator is fixed to a peripheral surface of the rotating body **46**. The rotating body **46** is not fixed to the main body **12**. Therefore, the rotating body **46**, the handle **50**, and the rotating ring **40** are rotatable with respect to the main body **12** and the first connecting surface **32**. When the restricting pin **58**, which will be described later, is in the predetermined restricting position, the rotation of the rotating ring **40** with respect to the main body **12** is restricted.

(17) FIG. 3 is a perspective view of the rotating ring **40**. As shown in FIG. 3, four engagement recesses **42** are formed in the rotating ring **40**. The engagement recess **42** is a substantially L-shaped notch that advances in the axial direction from the end of the rotating ring **40** (that is, the end closer to the first connector **10**) and then travels in the circumferential direction. Hereinafter, a portion of the engagement recess **42** that advances in the axial direction is referred to as an “inlet **42a**”, and a portion that advances in the circumferential direction is referred to as a “tail **42b**”.

(18) When the first connector **10** is brought close to the second connector **70** with the rotational phase of the engaging pin **76** (see FIG. 1) of the second connector **70** matched to the rotational phase of the inlet **42a**, the engaging pin **76** enters the interior of the inlet **42a**. As a result, the second connector **70** is drawn in the axial direction to the predetermined abutting position. In the abutting position, the second connecting surface **74** abuts against the first connecting surface **32**. When the second connector **70** is retracted to the abutting position and the rotating ring **40** is further rotated with respect to the first connecting surface **32**, the engaging pin **76** enters the tail **42b** and engages the tail **42b**. This engagement maintains the second connecting surface **74** in contact with the first connecting surface **32**.

(19) A receiving recess **44** is further formed in a part of the rotating ring **40** in the circumferential direction. The receiving recess **44** is configured by partially reducing the thickness of the peripheral wall of the rotating ring **40**. The receiving recess **44** receives an interlocking member **64**, which

will be described later.

(20) As shown in FIG. 2, the first valve **26** is connected to the interlocking member **64** and the first gear **62** via the rotation shaft **68**. An opening-closing lever **66** (see FIG. 1) is connected to an end portion of the first gear **62**. When the opening-closing lever **66** is swung, the first gear **62**, the interlocking member **64**, and the rotation shaft **68** are rotated, and the rotation shaft **68** is rotated, whereby the first valve **26** is opened and closed. That is, the first valve **26** is opened and closed by swinging the opening-closing lever **66**. When the first connector **10** is connected to the second connector **70**, the first gear **62** meshes with a second gear **84**, which will be described later.

(21) The interlocking member **64** is a non-circular member integrated with the first gear **62** on the upper surface of the first gear **62**. This interlocking member **64** suppresses improper opening of the first valve **26** and improper disconnection of the connector. The interlocking member **64** will be described in detail later.

(22) FIG. 4 and FIG. 5 are enlarged views of the section A of FIG. 2. As shown in FIGS. 4 and 5, the first connector **10** includes a pin hole **54**, a pressing pin **56**, a restricting pin **58**, and a spring **60**. The pin hole **54** is a hole elongated in the axial direction. The pin hole **54** extends over both the rotating body **46** and the flange portion **12a** of the main body **12**. The restricting pin **58** is disposed inside the pin hole **54**. The restricting pin **58** moves forward and backward between a restricting position for restricting the relative rotation of the rotating ring **40** with respect to the main body **12** and an allowable position for permitting the relative rotation.

(23) When the restricting pin **58** is in the restricting position, the restricting pin **58** straddles both the rotating body **46** and the main body **12**. FIG. 4 shows a state in which the restricting pin **58** is in the restricting position. When the restricting pin **58** is in the restricting position, the rotation of the rotating body **46** with respect to the main body **12** is restricted, and the rotation of the rotating ring **40** with respect to the first connecting surface **32** is restricted.

(24) When the restricting pin **58** is in the allowable position, the restricting pin **58** is completely inside the rotating body **46** and completely detached from the main body **12**. FIG. 5 illustrates a state in which the restricting pin **58** is in an allowable position. The spring **60** biases the restricting pin **58** in the restricting position direction. When the restricting pin **58** is in the allowable position, rotation of the rotating body **46** with respect to the main body **12** is permitted, and rotation of the rotating ring **40** with respect to the first connecting surface **32** is also permitted.

(25) The pressing pin **56** is a pin arranged on the axial end side with respect to the restricting pin **58**. The first connecting plate **30** is formed with a passage hole **35** through which the pressing pin **56** passes. The proximal end portion of the pressing pin **56** is inserted into the pin hole **54**, and the distal end portion of the pressing pin **56** is inserted into the passage hole **35**. As shown in FIG. 4, when the restricting pin **58** is in the restricting position, the distal end of the pressing pin **56** protrudes outward from the first connecting surface **32**. As shown in FIG. 5, when the second connecting surface **74** is pressed against the first connecting surface **32**, the pressing pin **56** and the restricting pin **58** are moved back to the axial proximal end side in a chain manner. As a result, the restricting pin **58** moves to the allowable position.

(26) Next, the second connector **70** will be described. As shown in FIG. 1, a second flow path **80** through which liquid hydrogen and hydrogen gas pass is formed inside the second connector **70**. A second valve **82** is provided in the second flow path **80**. A second gear **84** is connected to the second valve **82** via a rotation shaft (not shown). When the first connector **10** is coupled to the second connector **70**, the second gear **84** meshes with the first gear **62**. Then, the first valve **26** and the second valve **82** are synchronously opened and closed by swinging the opening-closing lever **66** in a state where the second gear **84** is meshed with the first gear **62**.

(27) A second connecting plate **72** is provided at the end of the second connector **70** (that is, at the end portion on the side of the first connector **10**). The end surface of the second connecting plate **72** functions as a second connecting surface **74** that abuts against the first connecting surface **32**. The four engaging pins **76** extend in the radial direction from the circumferential surface of the second

connecting plate **72**. The form of the engagement recess **42** and the engaging pin **76** may be changed. For example, the engagement recess **42** may be a groove instead of a notch. Further, the engaging pin **76** may be provided in the first connector **10**, the engagement recess **42** may be provided in the second connector **70**, respectively.

(28) Next, a procedure for connecting the first connector **10** and the second connector **70** will be described. When the connector is connected, the operator grasps the handle **50** so that the rotational phase of the inlet **42a** of the engagement recess **42** matches the rotational phase of the engaging pin **76**. If the rotational phase can be adjusted appropriately, the operator moves the first connector **10** axially and causes the engaging pin **76** to enter the inlet **42a**. As a result, the second connector **70** is axially retracted to the predetermined abutting position.

(29) When the second connector **70** is retracted to the abutting position, the second connecting surface **74** comes into contact with the first connecting surface **32**, and the pressing pin **56** is pressed in the proximal direction. As a result, the restricting pin **58** is retracted from the restricting position to the allowable position. When the restricting pin **58** reaches the allowable position, the rotation of the rotating ring **40** with respect to the first connecting surface **32** is permitted. In this state, the operator rotates the rotating ring **40** with respect to the first connecting surface **32**. As a result, the engaging pin **76** enters the tail **42b** of the engagement recess **42**. In this state, the movement of the second connector **70** in the axial direction with respect to the first connector **10** is restricted, so that the contact state between the first connecting surface **32** and the second connecting surface **74** is maintained. Then, in this state, the connection operation between the first connector **10** and the second connector **70** is completed.

(30) Here, as is obvious from the above description, in the present example, the rotation of the rotating ring **40** with respect to the main body **12** is restricted in a state where the second connecting surface **74** does not come into contact with the first connecting surface **32**. Further, in a state in which the second contact surface is in contact with the first connecting surface **32**, the rotation of the rotating ring **40** with respect to the main body **12** is allowed. The reason for such a configuration will be described.

(31) If the rotating ring **40** is always rotatable with respect to the main body **12**, the rotating ring **40** will swing and move relative to the main body **12**, making it difficult to align the rotational phase of the rotating ring **40** with respect to the second connector **70**. On the other hand, when the rotating ring **40** is not rotatable with respect to the main body at all times, the main body **12** itself needs to be rotated with the first connecting surface **32** abutting against the second connecting surface **74** in order to engage the engaging pin **76** in the tail **42b**. However, when the main body **12** is to be rotated, the first connecting plate **30** connected to the main body **12** is tilted or moved in the axial direction. Therefore, it is difficult to rotate the main body **12** while maintaining the contact state between the first connecting surface **32** and the second connecting surface **74**. Further, when the rotating ring **40** cannot rotate with respect to the main body **12**, the first connecting plate **30** rotates with respect to the second connecting plate **72** while being in contact with the second connecting plate **72**. In this case, the first connecting plate **30** and the second connecting plate **72** rub against each other, which leads to wear of the first connecting surface **32** and the second connecting surface **74**.

(32) On the other hand, in the case of the configuration in which the rotating ring **40** does not rotate with respect to the main body **12** when the first connecting surface **32** does not abut against the second connecting surface **74** as in the present example, the operation of positioning the rotation phase of the rotating ring **40** with respect to the second connector **70** is facilitated. Further, according to the present example, in a state in which the first connecting surface **32** is in close contact with the second connecting surface **74**, the rotating ring **40** can be rotated without rotating the main body **12** and the first connecting plate **30**. As a result, the second connector **70** can be engaged with the first connector **10** in a relatively simple operation.

(33) Next, the interlock function in the present example will be described. The connector

connection structure of the present example has an interlock function to suppress both the first valve **26** from being released at an improper timing and the connector from being released at an improper timing. This interlock function will be described referring to FIGS. **6A** to **7C** of the drawings.

(34) FIGS. **6A** to **7C** are schematic three-sided views of the rotating ring **40** and the interlocking member **64**. In FIGS. **6A** to **7C**, FIG. **6A** is a plan view of the rotating ring **40**. FIG. **7A** is a plan view of an interlocking member **64**. Further, FIG. **6B** is a rear view of the rotating ring **40**. **7B** is a rear view of the interlocking member **64**. Further, FIG. **6C** is a VIC-VIC cross-sectional view, and FIG. **7C** is a VIIC-VIIC cross-sectional view.

(35) FIGS. **6A** to **6C** show the first valve **26** in a closed state, and FIGS. **7A** to **7C** show the first valve **26** in an open state. As shown in FIGS. **6A** to **7C**, the interlocking member **64** is non-circular in plan view and is eccentric with respect to the rotation shaft **68**. The interlocking member **64** is positioned in a predetermined closed position when the first valve **26** is closed and is positioned in a predetermined open position when the first valve **26** is open. As shown in FIGS. **6A** to **6C**, when the first valve **26** is in the closed position, the entire interlocking member **64** is axially outward of the rotating ring **40**. Further, as shown in FIGS. **7A** to **7C**, when the rotating ring **40** is in the open position, the axial range of the interlocking member **64** partially overlaps with the axial range of the rotating ring **40**.

(36) Further, as described above, the rotating ring **40** rotates with respect to the first connecting plate **30**. Hereinafter, the rotational phase when the engaging pin **76** enters the inlet **42a** of the engagement recess **42** is referred to as an “initial rotational phase”, and the rotational phase when the engaging pin **76** advances to the end of the engagement recess **42** is referred to as a “connection rotational phase”. In the illustrated FIGS. **6A** to **6C**, the rotating ring **40** is in an initial rotational phase, and in the illustrated FIGS. **7A** to **7C**, the rotating ring **40** is in a connection rotational phase.

(37) As shown in FIGS. **6A** to **6C**, in the initial rotational phase, the receiving recess **44** of the rotating ring **40** is in a position not axially opposed to the interlocking member **64**. In this case, even if the first valve **26** is opened and the interlocking member **64** is moved to the open position, the interlocking member **64** interferes with the rotating ring **40**. In other words, in a state in which the connection between the first connector **10** and the second connector **70** is not completed, the interlocking member **64** interferes with the rotating ring **40**, and thus the first valve **26** cannot be opened.

(38) Further, as shown in FIGS. **7A** to **7C**, at the connection rotational phase, the receiving recess **44** of the rotating ring **40** is axially opposed to the interlocking member **64**. Here, as shown in FIGS. **7A** to **7C**, when the interlocking member **64** is moved to the open position, a portion of the interlocking member **64** is received in the receiving recess **44**. In other words, when the rotating ring **40** is in the connection rotational phase, the interlocking member **64** does not interfere with the rotating ring **40** even if the first valve **26** is opened. In other words, when the connection between the first connector **10** and the second connector **70** is completed, interference between the interlocking member **64** and the rotating ring **40** does not occur, so that the first valve **26** can be opened.

(39) In addition, when the interlocking member **64** is in the open position, even if the rotating ring **40** is to be rotated, the interlocking member **64** becomes obstructed, so that the rotating ring **40** cannot be rotated. Therefore, when the first valve **26** is opened, the rotating ring **40** is not moved from the connection rotational phase, and the connection between the first connector **10** and the second connector **70** cannot be released.

(40) As described above, by providing the interlocking member **64** and the receiving recess **44**, it is possible to reliably prevent the first valve **26** from being opened before the connection of the connector is completed, and the connection of the connector is released while the first valve **26** is opened. Further, as described above, in the present example, since the first gear **62** and the second gear **84** are engaged with each other, the first valve **26** and the second valve **82** are opened and

closed synchronously. With this configuration, it is possible to prevent the second valve 82 from being inappropriately opened and closed by the interlock function provided in the first connector 10. Thus, the safety of the liquid hydrogen supply operation can be further improved.

Claims

1. A connector connection structure comprising: a first connector for connection to one of a supply source and a reservoir of liquid hydrogen; and a second connector for connection to the other of the supply source and the reservoir, wherein: the first connector includes comprises: a main body, a first connecting surface that is fixed to the main body and abuts a second connecting surface of the second connector, a rotating ring that surrounds the first connecting surface and is rotatable with respect to the first connecting surface, and a restricting pin that prohibits a rotation of the rotating ring with respect to the first connecting surface; when the rotating ring takes a predetermined rotational phase with respect to the second connector, the rotating ring is able to pull in the second connector in an axial direction to an abutting position at which the second connecting surface abuts the first connecting surface, and further, rotates with respect to the first connecting surface in a state where the second connector is pulled in to the abutting position and partially engages with the second connector so as to maintain a connection state; and when the second connector is pulled in to the abutting position, the restricting pin moves from a restricting position at which the rotation of the rotating ring is prohibited to an allowable position at which the rotation of the rotating ring is allowed, wherein: when the first connector and the second connector are connected to each other, the rotating ring takes a predetermined connection rotational phase with respect to the first connecting surface; and the first connector further includes a first valve that opens and closes a flow path of the liquid hydrogen, and an interlocking member that prohibits opening of the first valve when the rotating ring does not take the connection rotational phase, and prohibits a rotation of the rotating ring from the connection rotational phase when the first valve is opened, wherein the interlocking member is non-circular and physically connected to the first valve.

2. The connector connection structure according to claim 1, wherein: the interlocking member is positioned at a closed position at which the interlocking member does not interfere with the rotating ring when the first valve is closed, and moves with the first valve so as to be positioned at an open position when the first valve is opened; the rotating ring is provided with a receiving recess in a portion in a circumferential direction; when the rotating ring does not take the connection rotational phase, the rotating ring interferes with the interlocking member in a process of moving from the closed position to the open position; when the rotating ring takes the connection rotational phase, the rotating ring receives at least part of the interlocking member positioned at the open position in the receiving recess; and when the interlocking member is partially received in the receiving recess, the rotation of the rotating ring with respect to the first connecting surface is restricted.

3. The connector connection structure according to claim 1, wherein: the first connector further includes an opening-closing lever that is mechanically connected to the first valve and opens and closes the first valve by being operated, and a first gear that rotates with the opening-closing lever; the second connector further includes: a second valve that opens and closes the flow path of the liquid hydrogen, and a second gear that rotates with the second valve and meshes with the first gear when the first connector is connected to the second connector; and when the first gear and the second gear mesh with each other, the first valve and the second valve are opened and closed in synchronization with each other.

4. A connector connection structure comprising: a first connector for connection to one of a supply source and a reservoir of liquid hydrogen; and a second connector for connection to the other of the supply source and the reservoir, wherein: the first connector comprises: a main body, a first connecting surface that is fixed to the main body and abuts a second connecting surface of the

second connector, a rotating ring that surrounds the first connecting surface and is rotatable with respect to the first connecting surface, and a restricting pin that prohibits a rotation of the rotating ring with respect to the first connecting surface; when the rotating ring takes a predetermined rotational phase with respect to the second connector, the rotating ring is able to pull in the second connector in an axial direction to an abutting position at which the second connecting surface abuts the first connecting surface, and further, rotates with respect to the first connecting surface in a state where the second connector is pulled in to the abutting position and partially engages with the second connector so as to maintain a connection state; and when the second connector is pulled in to the abutting position, the restricting pin moves from a restricting position at which the rotation of the rotating ring is prohibited to an allowable position at which the rotation of the rotating ring is allowed, wherein: the main body is provided with a flange portion that protrudes in a radial direction; the first connector further includes a rotating body that surrounds the main body and is rotatable together with the rotating ring with respect to the main body, a pin hole that extends in the axial direction so as to extend across the rotating body and the flange portion and into which the restricting pin is inserted, and a pressing pin of which a first part protrudes from the first connecting surface and a second part is inserted into the pin hole; the restricting position is a position at which the restricting pin extends in both of the flange portion and the rotating body, and the allowable position is a position at which the restricting pin is completely separated from the flange portion; and when the pressing pin is pressed in the axial direction by the second connecting surface, the pressing pin and the restricting pin are moved back in the axial direction in a chain manner, and the restricting pin moves from the restricting position to the allowable position.
